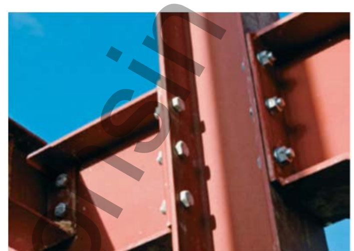
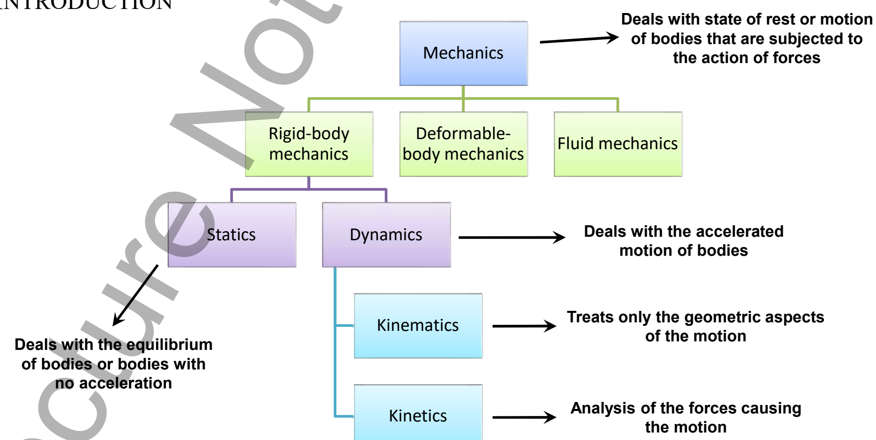
Mechanics Of Materials

- Mechanics Of Materials
- Loads
- Stress
- Strain



Mechanics Of Materials

Introduction



Essential for the design and analysis of many types of structural members, mechanical components, or electrical devices encountered in engineering

Mechanics Of Materials

INTRODUCTION

Fig.1 Flat Face Grips

Fig.2 Test Pieces

Mechanics Of Materials

APPLICATIONS

Millau Viaduct

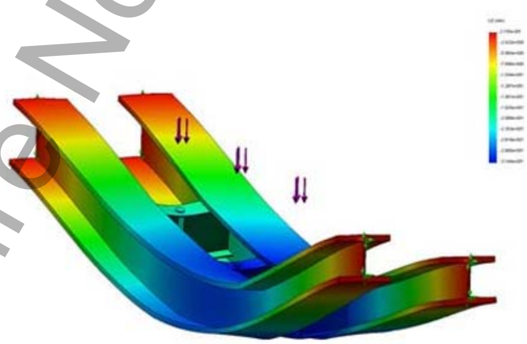
Eiffel Tower

Mechanics Of Materials

APPLICATIONS

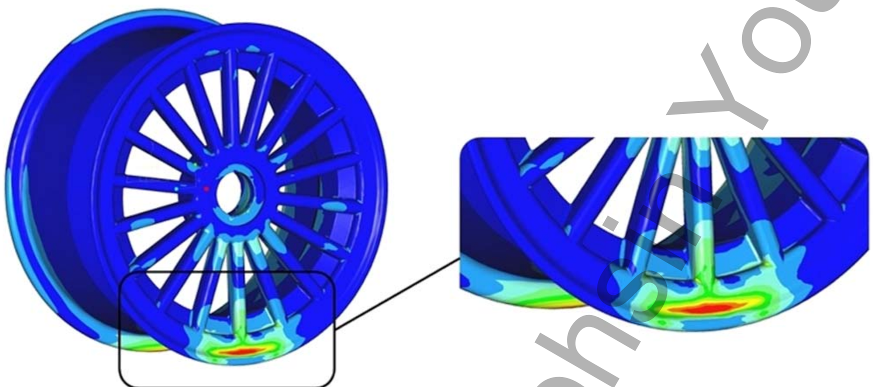
Mechanics Of Materials

DESIGN



Mechanics Of Materials

DESIGN



Mechanics Of Materials

TESTING

Mechanics Of Materials

Mistakes

Loads

External Loads

External Loads
- Surface Forces
  - Concentrated Load
  - Distributed Load
- Body Forces

Surface forces are caused by the direct contact of one body with the surface of another

Body force is developed when one body exerts a force on another body without direct physical contact

Idealizations

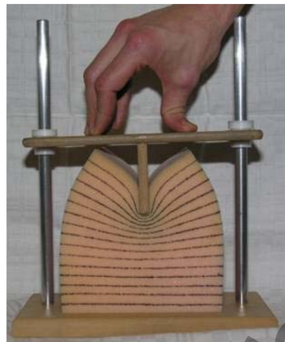
Concentrated Load → **Acts on a Point (Negligible Area)**

Distributed Load → **Acts on a Line (Narrow Strip)**

Loads

EXTERNAL LOADS

Sponge Model [1]



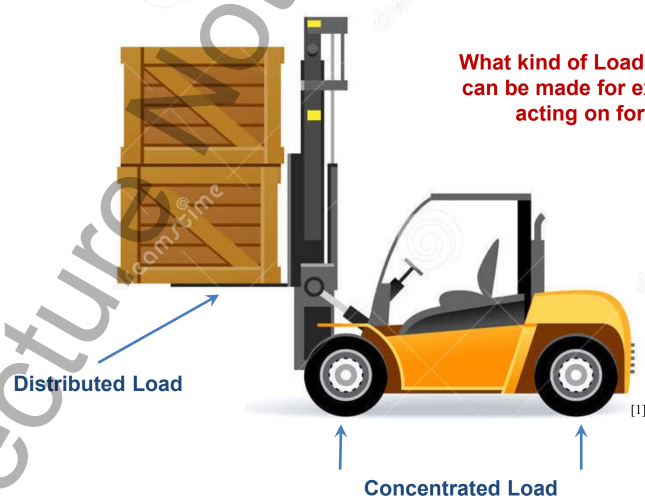
Concentrated Load

Distributed Load

[1] – Website: http://www.mace.manchester.ac.uk/project/teaching/civil/structuralconcepts/Statics/stress/stress_mod3.php Visited On: 10-05-2015

Loads

EXTERNAL LOADS

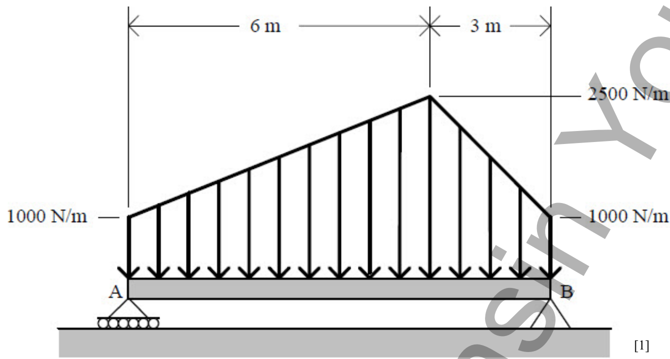


What kind of Load Idealizations can be made for external loads acting on fork lifter?

[1] – Website: http://www.dreamstime.com/royalty-free-stock-photography-forklift-image25095357 Visited On: 13-05-2015

Loads

External Loads - Distributed Loads



[1]

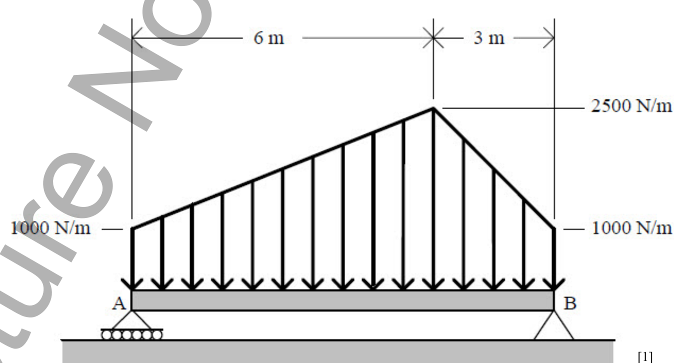
Resultant force is equivalent to the area under the distributed loading curve

Resultant acts through the centroid or geometric center of this area

[1] – Website: http://www.chegg.com/homework-help/questions-and-answers/beam-shown-bears-distributed-load-indicateddetermine-reactions-supports-b-q873787

Loads

External Loads - Distributed Loads



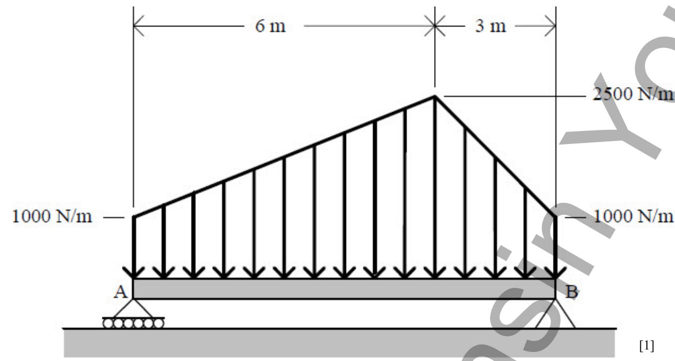
[1]

What is the Resultant Force Acting of the Bar?

[1] – Website: http://www.chegg.com/homework-help/questions-and-answers/beam-shown-bears-distributed-load-indicateddetermine-reactions-supports-b-q873787

Loads

External Loads - Distributed Loads



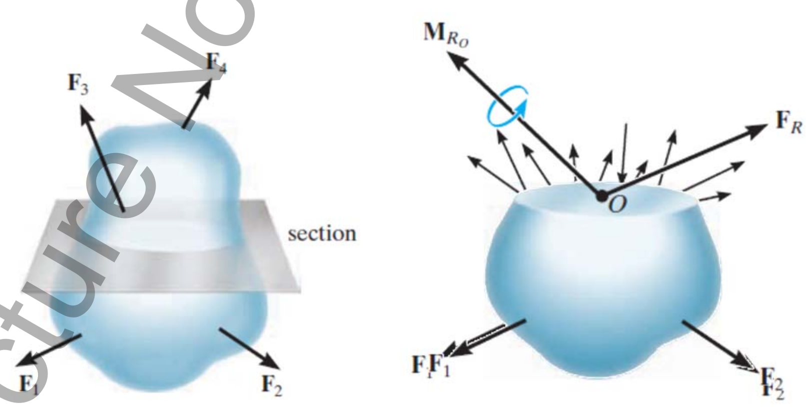
[1]

Where is this Resultant Acting?

[1] – Website: http://www.chegg.com/homework-help/questions-and-answers/beam-shown-bears-distributed-load-indicateddetermine-reactions-supports-b-q873787

Loads

Internal Resultant Loading

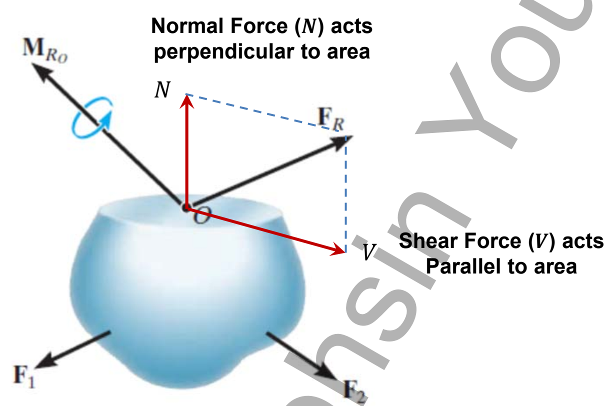


[1]

[1] - Russell C. Hibbeler. Mechanics of Materials 8th edition, Fig 1-2 (a,b,c), pp. 7

Loads

INTERNAL RESULTANT LOADING

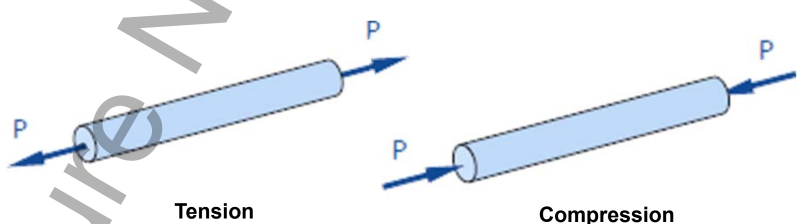


[1]

[1] - Russell C. Hibbeler. Mechanics of Materials 8th edition, Fig 1-2 (d), pp. 8

Loads

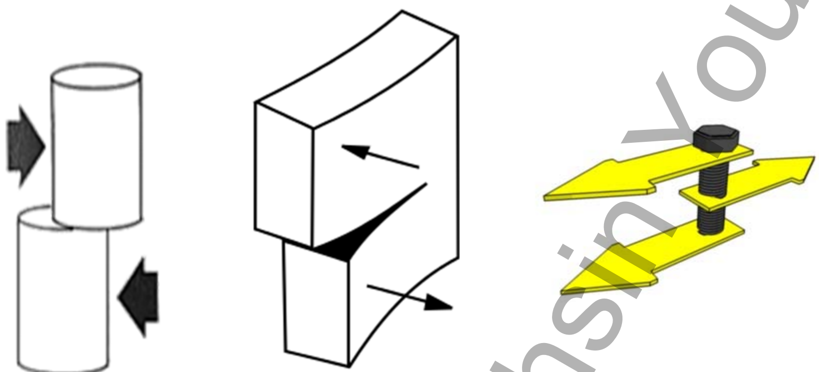
INTERNAL RESULTANT LOADING



Normal Force (N) acts perpendicular to area

Loads

INTERNAL RESULTANT LOADING

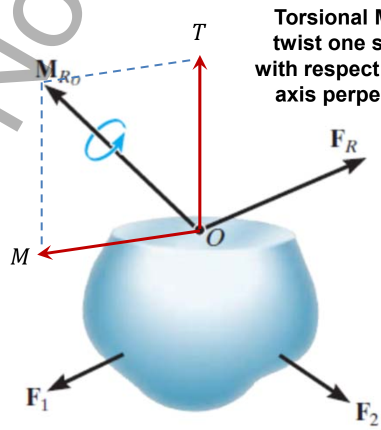


Shear Force (V) acts Parallel to area

Loads

INTERNAL RESULTANT LOADING

Torsional Moment (T) tends to twist one segment of the body with respect to the other about an axis perpendicular to the area



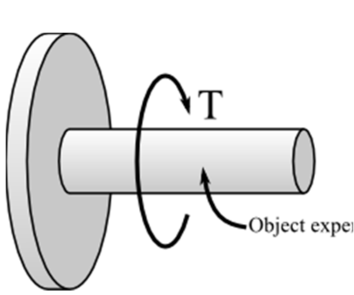
Bending Moment (M) tends to bend the body about an axis lying within the plane of the area.

[1]

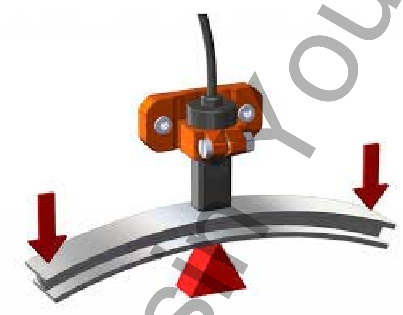
[1] - Russell C. Hibbeler. Mechanics of Materials 8th edition, Fig 1-2 (d), pp. 8

Loads

Internal Resultant Loading



Torsional Moment



Bending Moment

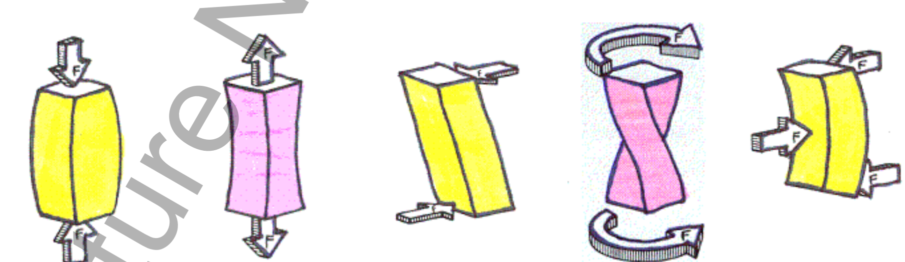
Torsional Moment (T) tends to twist one segment of the body with respect to the other about an axis perpendicular to the area

Bending Moment (M) tends to bend the body about an axis lying within the plane of the area.

Loads

Internal Resultant Loading

Identify Internal Loading

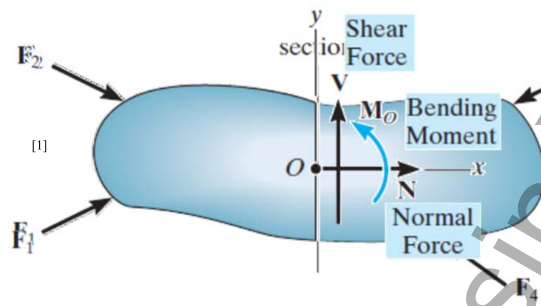


Compression | **Tension** | **Shear** | **Torsion** | **Bending**

[1] – Website: http://legacy.mos.org/discoverycenter/aotm/2012/04 Visited On: 01/05/2015

Loads

Coplanar Loadings



How Can One Calculate V, N and M_o?

$\sum \vec{F} = 0 \longrightarrow N$ and V

$\sum \vec{M_O} = 0 \longrightarrow M$

[1] - Russell C. Hibbeler. Mechanics of Materials 8th edition, Fig 1-3, pp. 9